

Computer Networks: Transport Layer Protocols

Transmission Control Protocol (TCP) and User Datagram Protocol (UDP)

TCP provides reliable, ordered, and error-checked delivery of a stream of octets.

TCP uses a three-way handshake: SYN, SYN-ACK, and ACK to establish a connection.

Flow control is managed by sliding windows, and congestion control uses AIMD.

UDP is connectionless and lightweight, used for DNS, VoIP, and video streaming.